Problem 1: Betting on the 2020 Election

Setup

By Savage's framework, Victor chooses $a^* = \arg\max_a \mathbb{E}_{X\sim P}[U(a, X)]$. Here the state X is whether Biden wins, Victor's belief is $P(\text{Biden wins}) = 0.85$, the action is the bet size $b \in [0, d]$, and U is applied to his final wealth.

At a market price of 0.6, a \$0.6 contract pays \$1 if Biden wins. Betting b buys $b/0.6$ contracts, so Victor gains $b/0.6 - b = \frac{2}{3}b$ if Biden wins and loses b otherwise:

$$W(b) = \begin{cases} d + \frac{2}{3}\,b & \text{with probability } 0.85, \\ d - b & \text{with probability } 0.15. \end{cases}$$

(a) Linear utility

Claim: with $U(y) = y$, Victor bets his entire bankroll, $b^* = d$. The expected utility is

$$\begin{aligned} \mathbb{E}[U(W(b))] &= 0.85\left(d + \tfrac{2}{3}\,b\right) + 0.15\,(d - b) \\ &= d + \left(0.85 \cdot \tfrac{2}{3} - 0.15\right) b \\ &= d + \tfrac{5}{12}\,b. \end{aligned}$$

This is a linear function of b with strictly positive slope $\frac{5}{12}$. A linear function with positive slope on the closed interval $[0, d]$ attains its maximum at the right endpoint. Therefore $b^* = d$. □

Does linear utility make sense? No. Betting everything gives a 15% chance of ending with *zero* wealth, and linear utility treats this as merely "d fewer dollars," exactly offset in value by an equal-sized gain. This conflicts with how people value money.

(b) Log utility

With $U(y) = \log y$, Victor maximizes

$$g(b) := \mathbb{E}[\log W(b)] = 0.85 \log\left(d + \tfrac{2}{3}\,b\right) + 0.15 \log(d - b), \qquad 0 \le b < d.$$

First-order condition. By the chain rule,

$$g'(b) = \frac{0.85 \cdot \frac{2}{3}}{d + \frac{2}{3}\,b} - \frac{0.15}{d - b} = \frac{17/30}{d + \frac{2}{3}\,b} - \frac{3/20}{d - b}.$$

Setting $g'(b) = 0$ and cross-multiplying:

$$\begin{aligned}
\tfrac{17}{30}\,(d-b) &= \tfrac{3}{20}\left(d+\tfrac{2}{3}\,b\right) \\
34\,(d-b) &= 9\left(d+\tfrac{2}{3}\,b\right) \qquad \text{(multiply both sides by 60)} \\
34d - 34b &= 9d + 6b \\
25d &= 40b,
\end{aligned}$$

so

$$\boxed{b^* = \tfrac{5}{8}\,d = 0.625\,d.}$$

This is a maximum. The second derivative is

$$g''(b) = -\frac{0.85\left(\frac{2}{3}\right)^2}{\left(d+\frac{2}{3}\,b\right)^2} - \frac{0.15}{(d-b)^2} < 0,$$

so g is strictly concave and the stationary point is its unique global maximum. Also $b^* = \frac{5}{8}d \in [0, d)$ is feasible. As $b \to d$, $\log(d-b) \to -\infty$, so going all-in is now infinitely bad, the opposite of part (a).

Outcomes.

- Biden wins: $W = d + \frac{2}{3} \cdot \frac{5}{8}\,d = \frac{17}{12}\,d \approx 1.42\,d.$
- Biden loses: $W = d - \frac{5}{8}\,d = \frac{3}{8}\,d = 0.375\,d.$

(c) Would I make this bet with a net worth of $500,000?

The bet from part (b) would be $0.625 \times \$500{,}000 = \$312{,}500$. I would not bet this much, for the following reasons.

1. My utility is more risk-averse than log**.** My utility decreases faster than $\log y$ as wealth approaches zero. In Savage's framework this is perfectly rational: with a more concave U (e.g. $U(y) = y^{1-\gamma}/(1-\gamma)$ with $\gamma > 1$, where $\gamma = 1$ corresponds to log), the optimal bet is smaller than $0.625\,d$.

2. I am not confident that $p = 0.85$ is correct. Savage's theorem says what to do *given* a belief P, but not whether P is right. The market price of 0.6 is itself strong evidence: it is an equilibrium set by many participants with real money at stake. The calculation in (b) assumes with certainty that the forecast is right and the market is wrong.

Conclusion. I might place a small bet if I trusted the model's edge, but nowhere near $312,500. The decision rule is only as good as the belief P and utility U fed into it, and here I doubt both the belief and the appropriateness of log utility for me.